

CS549 Final

I. QUESTION: NEURAL NETWORKS

Draw a single input, single hidden layer (with a single hidden node) neural network (multilayer perceptron) for two-class classification problem. Derive the update equation for

- the weight between the input unit and hidden layer
- the weight between the hidden unit and the output.

Do not forget the bias unit.

II. QUESTION: VECTOR SEMANTICS

Given a corpus consisting of N documents, and a query of Q , write an algorithm which finds the nearest document D to the given query Q .

III. QUESTION: WORDNET

Given two noun synsets S_1 and S_2 in the Turkish wordnet, propose an algorithm to find the shortest distance in the hypernym tree between S_1 and S_2 .

IV. QUESTION: SEMANTIC ROLE LABELING

agree.01

Arg0: Agreeer

Arg1: Proposition

Arg2: Other entity agreeing

fall.01

Arg1: Logical subject, patient, thing falling

Arg2: Extent, amount fallen

Arg3: start point

Arg4: end point, end state of arg1

increase.01

Arg0: causer of increase

Arg1: thing increasing

Arg2: amount increased by, EXT, or MNR

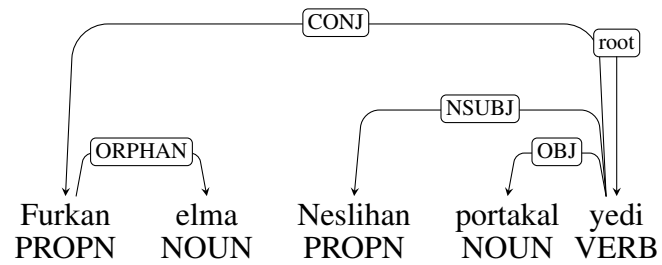
Arg3: start point

Arg4: end point

Annotate the following sentences with the arguments above:

- The group agreed it wouldn't make an offer
- Sales fell to \$25 million from \$27 million
- John agrees with Mary on everything.
- The price of bananas was increased again by Big Fruit Co.
- The average junk bond fell by 4.2%

V. QUESTION: DEPENDENCY PARSING



Construct the trace of the given transition-based parse above. Show stack, word list, action and relation added at each iteration. Assume that arc-standard parsing has been used.

VI. QUESTION: SYNTACTIC PARSING

Given the following grammar L_2 in CNF form:

$$\begin{aligned}
 S &\rightarrow NP VP \\
 NP &\rightarrow Det Nominal \\
 Nominal &\rightarrow Noun Noun \\
 VP &\rightarrow Verb NP \\
 VP &\rightarrow Verb PP \\
 VP &\rightarrow VP PP \\
 PP &\rightarrow Prep NP \\
 Noun &\rightarrow flight, trip, super, morning \\
 Verb &\rightarrow is \\
 Det &\rightarrow the, this \\
 Prep &\rightarrow on
 \end{aligned}$$

Parse the sentence "This super trip is on the morning flight" using CKY parsing. Show your work.